

**Smart Health Care
DISITI
Venue (Mansar Building)
(Room No: 12LH1003)
(16/07/2025)**

Day	Time	Paper Id	Name
16th July, 2025	11:00 am to 11:15 am	<i>Session Introduction</i>	
	11:15 am to 11: 30 am	124	AI-Driven Dermatology: Enhancing Melanoma Detection with XAI and Deep Learning
	11:30 am to 11:45 am	125	Explainable Skin cancer Detection via Hybrid CNN and Adaptive Post-Hoc Explanations
	11:45 am to 12:00 pm	188	Detect Before Decay: Early Diagnosis of Dental Caries via Intraoral Scans
	12:00 pm to 12:15 pm	223	Explainability-Driven Multimodal Glioma Segmentation Using a Hybrid CNN-Transformer Architecture and IoU-Based Attribution Analysis
	12:15 pm to 12:30 pm	226	Graph-Based Classification of Lung Cancer Histopathology Using SimCLR Feature Extraction and Graph Attention Networks
	12:30 pm to 12:45 pm	263	Brain Tumor Extraction using Adaptive Non convex and Superpixel Motion Detection

Medical Diagnostics and E-Healthcares**D1SIT2****Venue (Mansar Building)****(Room: 12LH1013)****(16/07/2025)**

Day	Time	Paper Id	Name
16th July, 2025	11:00 am to 11:15 am	<i>Session Introduction</i>	
	11:15 am to 11: 30 am	34	HybridMorph: Towards usage of synthetic with real data for medical MR images registration
	11:30 am to 11:45 am	39	Glaucoma Detection in Retinal Images via Attention-Based Hybrid Deep Model with Fibonacci-Inspired Ring Features
	11:45 am to 12:00 pm	47	Whirling Triangle-Based Geometric and Texture Feature Extraction with Attention-Driven Hybrid Models for Multiclass Osteoporosis Classification from Knee X-rays
	12:00 pm to 12:15 pm	51	Leveraging DeepLabV3+ with Explainable AI for Brain Tumor Segmentation in 2D MRI
	12:15 pm to 12:30 pm	66	Impact of Age and Gender on the Accuracy of Computational Model-Based Heart Disease Prediction
	12:30 pm to 12:45 pm	96	A Secure Collaborative Approach for Autism Spectrum Disorder Detection Using Decentralized Deep Learning

Industry Session
Venue (Mansar Building)
(Room: 12LH2004)
(16/07/2025)

Day	Time	Industry
16th July, 2025	2:00 pm to 2:30 pm	CPMU, AICOE (Talk by : Dr. Soma Dhaval)
	2:30 pm to 3:00 pm	iKITES Technologies Pvt. Ltd. (Talk by: Dipinder Sekhon / Vishal Singh)
	3:00 pm to 3:30 pm	Rohde & Schwarz

WORKSHOP
AI-Driven Biometrics for Forensic Intelligence: Trends, Challenges, and Applications
Venue (Mansar Building)
(Room: 12LH2003)
(16/07/2025)

Day	Time	Paper Id	Name
16th July, 2025	2:00 pm to 2:15 pm	<i>Session Introduction</i>	
	2:15 pm to 2:30 pm	154	Enhancing UPI Transactions Fraud Detection with Machine Learning and Biometric Authentication
	2:30 pm to 2:45 pm	112	A Review on AI-Based Cognitive Systems for Digital Wellbeing

Science Behind the Systems
D1S3T1
Venue (Mansar Building)
(Room: 12LH1004)
(16/07/2025)

Day	Time	Paper Id	Name
16th July, 2025	4:00 pm to 4:15 pm	<i>Session Introduction</i>	
	4:45 pm to 5:00 pm	23	Deep Convolutional Transform Learning Regularized Transductive Inversion
	5:00 pm to 5:15 pm	251	MobileNet-V2 + EfficientNet-B0 and DenseNet-121 + ResNet-50 Late-Fusion-Based Hybrid Deep Learning Architectures for Malware Detection
	5:15 pm to 5:30 pm	145	Boosting MobileNet Performance on Embedded Systems Using SE Modules
	5:30 pm to 5:45 pm	186	Fake News Detection Using Cellular Automata And Deep Learning
	5:45 pm to 6:00 pm	267	Quantum Convolutional Neural Networks with Residual Learning: Advancing Quantum Model Architectures
	6:00 pm to 6:15 pm	225	Hybrid Deep Learning and Explainable AI Approach Towards Trustworthy EV Stock Predictions

Intelligent Systems in Practice**D1S3T2****Venue (Mansar Building)****(Room: 12LH1014)****(16/07/2025)**

Day	Time	Paper Id	Name
16th July, 2025	4:30 am to 4:45 am	<i>Session Introduction</i>	
	4:45 pm to 5:00 pm	50	Attention Based Multi-patch Hierarchical Network with Non-local Information for Smartphone Image Denoising
	5:00 pm to 5:15 pm	46	Explainable Parallel CNN Model with Dense-Squeeze and Excitation Blocks for Corn Leaf Disease Classification
	5:15 pm to 5:30 pm	135	UnTherISR: Unsupervised Approach for Thermal Image Super-Resolution
	5:30 pm to 5:45 pm	55	Extractive Text Summarization using BERT with Weighted Pooling
	5:45 pm to 6:00 pm	136	MAMBA-TBPS: An Efficient Attribute-Relation-Sensitive Framework for Text-Based Person Search
	6:00 pm to 6:15 pm	162	ContextualVQA: An Efficient Ambient Assisted Monitoring framework Using Visual Question Answering

Emerging Trends in Biometrics and Other Applications

D2S4TI

Venue (Mansar Building)

(Room No: 12LH1003)

(17/07/2025)

Day	Time	Paper Id	Name
17th July, 2025	11:00 am to 11:15 am	<i>Session Introduction</i>	
	11:15 am to 11:30 am	33	Towards Realistic Synthetic Palmprint Data: A Smartphone Video Dataset with Diverse Variations
	11:30 am to 11:45 am	228	Deep Liveness: Face Liveness Detection Using a Lightweight U-Net-Based Deep Architecture
	11:45 am to 12:00 pm	236	Enhancing Latent Fingerprint Quality Assessment Using a Deep Learning Pipeline
	12:00 pm to 12:15 pm	252	Adversarial Representation Learning for Presentation Attack Detection on Face Recognition Systems
	12:15 pm to 12:30 pm	284	Sensei: Assistive Real-Time Perception for Visually Impaired Users with Depth-Aware Object and Face Recognition
	12:30 pm to 12:45 pm	13	Facial Action Unit Analysis for Indian Classical Dance Forms: A Micro-Emotion Sequencing Framework Approach

Intelligent Surveillance Systems**D2S4T2****Venue (Mansar Building)****(Room: 12LH1013)****(17/07/2025)**

Day	Time	Paper Id	Name
17th July, 2025	11:00 am to 11:15 am	<i>Session Introduction</i>	
	11:15 am to 11: 30 am	265	Deep Image Prior-Based Segmentation: A Study and proposal of Deeper U-Net
	11:30 am to 11:45 am	22	A Lightweight System for Attention-based Night-time Traffic Accident Detection
	11:45 am to 12:00 pm	246	High-Precision Traffic Accident Detection Using YOLOv11 Model and Image Processing with Deep Learning Techniques
	12:00 pm to 12:15 pm	75	Weighted TimeSformer Multi-Region CNN Network for Personalized and Context-aware Activity Localization in Ambient Assisted Living
	12:15 pm to 12:30 pm	222	Dual-modal Fusion Deep Learning Network for Traffic Targets Detection of Autonomous Vehicles
	12:30 pm to 12:45 pm	238	Real-Time Combat Training Analytics: Skeleton-based Temporal Action Localization in Unstructured Video

Industry Session
Venue (Mansar Building)
(Room: 12LH2004)
(17/07/2025)

Day	Time	Industry
17th July, 2025	2:00 pm to 2:30 pm	<i>Innomatics</i>
	2:30 pm to 3:00 pm	<i>Masai School</i>

WORKSHOP
Advances in Underwater Surveillance: Technologies, Challenges, and Future Directions
(AUSTech 2025)

Venue (Mansar Building)
(Room: 12LH3010)
(17/07/2025)

Day	Time	Paper Id	Name
17th July, 2025	2:00 pm to 2:45 pm	<i>Plenary talk by Prof. Santanu Chaudhury</i>	
	2:45 pm to 3:00 pm	131	Deep Learning Approach based Object Detection in Underwater SONAR Images
	3:00 pm to 3:15 pm	179	A Human Vision Neuroscience-Driven Deep Neural Framework for Change Detection in Underwater Scenes
	3:15 pm to 3:30 pm	253	AUSREGSE: Advanced Underwater Segmentation with ResNet50 with enhancement-guided Squeeze and Excitation Blocks
	3:30 pm to 3:45 pm	221	A Quality-Guided Framework for Low-Light Underwater Image Enhancement Using Degradation Classification and Benchmark Evaluation
	3:45 pm to 4:00 pm	237	Vision-Based Underwater Docking Station Detection for Autonomous Underwater Vehicles

Green Intelligence
D2S6T1
Venue (Mansar Building)
(Room: 12LH1004)
(17/07/2025)

Day	Time	Paper Id	Name
17th July, 2025	4:00 pm to 4:15 pm	<i>Session Introduction</i>	
	4:45 pm to 5:00 pm	29	A Vision-Guided Autonomous Weeding Robot using Image-Based Weed Classification for Sustainable Agriculture
	5:00 pm to 5:15 pm	44	A comparative analysis of different models for crop price forecasting
	5:15 pm to 5:30 pm	110	Precision Soil Monitoring via UAV-Captured RGB Imaging and Spectral Band Regression Using Deep Learning
	5:30 pm to 5:45 pm	202	An Explainable Deep Learning Framework for Banana Leaf Disease Detection and Analysis
	5:45 pm to 6:00 pm	286	Stacked Ensemble Learning for Global Crop Yield Prediction: A Comparative Study of Machine Learning Techniques
	6:00 pm to 6:15 pm	94	SAPxAl: Deep Features with Explainable Artificial Intelligence Techniques for Prediction of Saffron Adulteration

**Remote Sensing and Surveillance
D2S6T2**

**Venue (Mansar Building)
(Room: 12LH1014)
(17/07/2025)**

Day	Time	Paper Id	Name
17th July, 2025	4:30 am to 4:45 am	<i>Session Introduction</i>	
	4:45 pm to 5:00 pm	262	Saliency Guided Tile -Based Image Segmentation
	5:00 pm to 5:15 pm	122	PinTags: A Visual Fiducial Marker System for Logistics
	5:15 pm to 5:30 pm	227	WaterIDViT: Open Water Body Extraction in High-resolution Remote Sensing Images Using Vision Transformers
	5:30 pm to 5:45 pm	254	Improvisation of Yolov4 Algorithm for Multiple Object Tracking Using Kalman Filtering Hungarian Method
	5:45 pm to 6:00 pm	118	Automated Georeferencing of Topographic Maps via OCR and In-Context Multimodal LLM Reasoning
	6:00 pm to 6:15 pm	133	PRISM: Progressive Regional Integration with Synergetic Multi-Attention Transformer for Single Image Super-Resolution

Multimodal Signal Analysis and Surveillance
D3S7TI

Venue (Mansar Building)
(Room No: 12LH1003)
(18/07/2025)

Day	Time	Paper Id	Name
18th July, 2025	11:00 am to 11:15 am	<i>Session Introduction</i>	
	11:15 am to 11: 30 am	38	Blind Spot Aware and Speed-Sensitive Emergency Vehicle Detection Using YOLO and Faster R-CNN Synergy
	11:30 am to 11:45 am	219	Demystifying Transformer: A Case Study on Graph-Based Interpretability and Attention Layer Dynamics
	11:45 am to 12:00 pm	183	Class-Agnostic Triple Attention Model Towards Optimization and Variable Rate Compression in Wildlife Camera Trap Images
	12:00 pm to 12:15 pm	184	Device-generalized representations for acoustic scene classification
	12:15 pm to 12:30 pm	272	Enhancing LSTM-based Piano Music Generation with Self-Attention on the Lakh MIDI Dataset
	12:30 pm to 12:45 pm	83	Prediction of Direction-of-Arrival and Distance for Binaural Sound Sources Using a CRNN-Based Model

E-Healthcares D3S7T2 Venue (Mansar Building) (Room: 12LH1013) (18/07/2025)			
Day	Time	Paper Id	Name
18th July, 2025	11:00 am to 11:15 am	<i>Session Introduction</i>	
	11:15 am to 11: 30 am	277	Robust Spatial Enhancement of ASL-MRI based on 3DFSRCNN Framework
	11:30 am to 11:45 am	291	Protein Methylation Site Prediction based on spatial feature of Amino Acids using SMOTE and Stratified Cross-Validation
	11:45 am to 12:00 pm	261	A Classifier for Spinal Tumors using Novel Data Augmentation
	12:00 pm to 12:15 pm	242	A Proximal Approach to Image Deconvolution under Mixed Poisson-Gaussian Noise
	12:15 pm to 12:30 pm	111	Emotion Recognition with EEG Signals using Transformer-based Trainable Adjacency Matrix and Graph Convolutional Network
	12:30 pm to 12:45 pm	280	Sign Language Translation Using Multi-Modal Transformer Network

WORKSHOP
3rd Workshop on High Throughput Vision Based Phenotyping
(HTVP'25)
Venue (Mansar Building)
(Room: 12LH2003)
(18/07/2025)

Day	Time	Paper Id	Name
18th July, 2025	2:00 pm to 2:15 pm	<i>Session Introduction</i>	
	2:15 pm to 2:30 pm	20	Semi-Supervised Approach for Disease Severity
	2:30 pm to 2:45 pm	132	Vegetation region localization for plant in field having specular reflection
	2:45 pm to 3:00 pm	153	Quantifying Plant Structural Complexity Using an Image-Based Structural Complexity Index (SCI)
	3:00 pm to 3:15 pm	276	SRSegFormer: Joint Super-Resolution and Segmentation Framework for Fine-Scale Root Phenotyping